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1 #Taylor Swift Playlist Generator: Generates a song based on user input OR
2 a playlist based on a certain place or mood OR generates a mini Eras Tour
3 setlist with one surprise song
4 #Initialization
5 import pandas as pd
6 import random
7 data = pd.read_csv('ts.csv')
8 name = data['Song Title'].tolist()
9 album = data['Album'].tolist()
10 length = data['Length'].tolist()
11 season = data['Season'].tolist()
12 location = data['Location'].tolist()
13 mood = data['Mood'].tolist()
14 collab = data['Collab'].tolist()
15 new = data['New'].tolist()
16 popularity = data['Popularity'].tolist()
17 setlist = data['Eras Tour Setlist'].tolist()
18 #Functions
19 #Choosing popularity of songs, based on a 1-10 scale, 1 being very
20 unknown, 10 being very popular
21 popsongs = []
22 def pop():
23     userp = input("Would you like your song to be popular or niche? ")
24     if userp == "popular" or userp == "Popular":
25         for i in range(len(name)):
26             if popularity[i] >= 6:
27                 popsongs.append(name[i] + " from the album " + album[i])
28     elif userp == "niche" or userp == "Niche":
29         for i in range(len(name)):
30             if popularity[i] <= 5:
31                 popsongs.append(name[i] + " from the album " + album[i])
32     else:
33         print("Please try again!")
34         pop()
35 #Season that song gives vibe of
36 season_song = []

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37 season_album = []
38 def seasons():
39     userseas = input("What season would you like your song to feel like?
40 ")
41     if userseas == "Fall" or userseas == "fall":
42         for i in range(len(name)):
43             if season[i] == "Fall":
44                 season_song.append(name[i] + " from the album " +
45 album[i])
46     elif userseas == "Winter" or userseas == "winter":
47         for i in range(len(name)):
48             if season[i] == "Winter":
49                 season_song.append(name[i] + " from the album " +
50 album[i])
51     elif userseas == "Spring" or userseas == "spring":
52         for i in range(len(name)):
53             if season[i] == "Spring":
54                 season_song.append(name[i] + " from the album " +
55 album[i])
56     elif userseas == "Summer" or userseas == "summer":
57         for i in range(len(name)):
58             if season[i] == "Summer":
59                 season_song.append(name[i] + " from the album " +
60 album[i])
61     else:
62         print("Error. Not a correct season")
63         seasons()
64 #Length of songs
65 length_song = []
66 def length_gen():
67     userlen = input("Would you like the song to be short (<3.5) or long
68 (3.5+)? ")
69     if userlen == "short" or userlen == "Short":
70         for i in range(len(length)):
71             if length[i]<3.5:

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72         length_song.append(name[i] + " from the album " +
73 album[i])
74     elif userlen == "long" or userlen == "Long":
75         for i in range(len(length)):
76             if length[i] >= 3.5:
77                 length_song.append(name[i] + " from the album " +
78 album[i])
79     else:
80         print("Please enter a valid length.")
81         length_gen()
82 #Locations/purpose of playlist
83 location_playlist = []
84 def locations():
85     userloco = input("Would you like a playlist for Driving, Beach,
86 Studying, Train, or Getting Ready? ")
87     if userloco == "Driving" or userloco == "driving":
88         for i in range(len(name)):
89             if location[i] == "Driving":
90                 location_playlist.append(name[i])
91     elif userloco == "Beach" or userloco == "beach":
92         for i in range(len(name)):
93             if location[i] == "Beach":
94                 location_playlist.append(name[i])
95     elif userloco == "Studying" or userloco == "studying":
96         for i in range(len(name)):
97             if location[i] == "Studying":
98                 location_playlist.append(name[i])
99     elif userloco == "Train" or userloco == "train":
100         for i in range(len(name)):
101             if location[i] == "Train":
102                 location_playlist.append(name[i])
103     elif userloco == "Getting Ready" or userloco == "getting ready":
104         for i in range(len(name)):
105             if location[i] == "Getting Ready":
106                 location_playlist.append(name[i])
107     else:

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108         print("Please enter a valid playlist purpose/location.")
109         return
110 #Mood of playlist
111 playlist_mood = []
112 def moods():
113     usermood = input("What would you like the mood of the playlist to
114 be?
115
116         Happy
117         Sentimental
118         Sad
119         Anger
120         Love
121         Yearning
122         Payback
123         """)
124 if usermood == "Happy" or usermood == "happy":
125     for i in range(len(name)):
126         if mood[i] == "Happy":
127             playlist_mood.append(name[i])
128 elif usermood == "Sentimental" or usermood == "sentimental":
129     for i in range(len(name)):
130         if mood[i] == "Sentimental":
131             playlist_mood.append(name[i])
132 elif usermood == "Sad" or usermood == "sad":
133     for i in range(len(name)):
134         if mood[i] == "Sad":
135             playlist_mood.append(name[i])
136 elif usermood == "Anger" or usermood == "anger":
137     for i in range(len(name)):
138         if mood[i] == "Anger":
139             playlist_mood.append(name[i])
140 elif usermood == "Love" or usermood == "love":
141     for i in range(len(name)):
142         if mood[i] == "Love":
143             playlist_mood.append(name[i])
144 elif usermood == "yearning" or usermood == "Yearning":

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144         for i in range(len(name)):
145             if mood[i] == "Yearning":
146                 playlist_mood.append(name[i])
147     elif usermood == "Payback" or usermood == "payback":
148         for i in range(len(name)):
149             if mood[i] == "Payback":
150                 playlist_mood.append(name[i])
151     else:
152         print("Error! Please enter a valid letter.")
153     return
154 #Deciding factor of playlist
155 def playlist_decision():
156     decide = input("Would you like a playlist based around a location or a
157 mood? ")
158     if decide == "location" or decide == "Location":
159         locations()
160         print(location_playlist)
161     elif decide == "mood" or decide == "Mood":
162         moods()
163         print(playlist_mood)
164 #Mini Eras Tour setlist generator
165 eras_t = []
166 user_setlist = []
167 def eras_tour(quantity):
168     for i in range(len(name)):
169         if setlist[i] == "Yes":
170             eras_t.append(name[i])
171     for i in range(quantity):
172         user_setlist.append(random.choice(eras_t))
173     print(user_setlist)
174 #Randomly generated surprise song (Taylor performed two every show from
175 songs not in the setlist)
176 surprise_songs = []
177 def surp_songs():
178     for i in range(len(name)):
179         if setlist[i] == "No":

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180         surprise_songs.append(name[i])
181     print("...and your surprise song is: " + random.choice(surprise_songs))
182 #Process of getting a song
183 #Finding intersections of arrays with algorithm from
184 https://flexiple.com/python/intersection-two-list-python
185 def song():
186     pop()
187     seasons()
188     length_gen()
189     common_items = list(set(popsongs).intersection(season_song))
190     songs_criteria = list(set(length_song).intersection(common_items))
191     print("Your random song is...." +
192         random.choice(songs_criteria))
193 #Main menu
194 def menu():
195     print("I was enchanted to generate some songs for you!")
196     print("Welcome to the Taylor Swift generator.")
197     user_type = input("Would you like to generate a playlist, a random
198 song, or find your mini ERAS TOUR setlist? ")
199     if user_type == "Playlist" or user_type == "playlist":
200         playlist_decision()
201     if user_type == "Random song" or user_type == "Random Song" or
202 user_type == "random song" or user_type == "song" or user_type == "Song":
203         song()
204     elif user_type == "ERAS TOUR" or user_type == "Eras Tour" or user_type
205 == "eras tour" or user_type == "Eras tour" or user_type == "setlist" or
206 user_type == "Setlist" or user_type == "eras tour setlist" or user_type ==
207 "ERAS TOUR setlist" or user_type == "Eras Tour Setlist" or user_type ==
208 "Eras tour setlist" or user_type == "mini eras tour setlist" or user_type
209 == "mini ERAS TOUR setlist" or user_type == "mini eras tour" or user_type
210 == "mini ERAS TOUR":
211         eras_tour(3)
212         surp_songs()
213     else:
214         print("Please enter a valid choice.")
215     menu()

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217 menu ( )